

simTIM

A Temporal Cybersecurity Simulator

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ARES 2026

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2026-08-21

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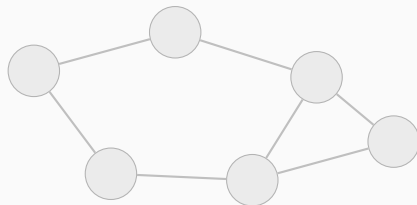
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Why model cybersecurity?



Which servers are vulnerable and need hardening?

How should we change the network topology?

Simulations provide **what-if analysis**.

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└ Why model cybersecurity?

0:30–1:05. Put them in the chair, do not read the questions verbatim.

Imagine you are a security analyst, or a network planner, at an organisation. You have been given a budget and the task of improving security in the company network. Which servers are particularly vulnerable and need hardening? How do you optimise the topology?

What you would really like is what-if analysis — and that is why there has been so much research into modelling cybersecurity.

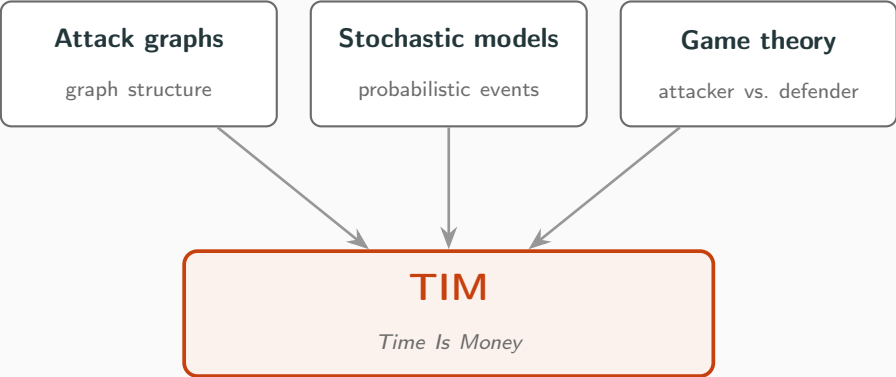
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TIM: three families of security models

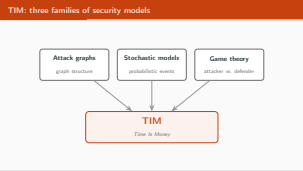


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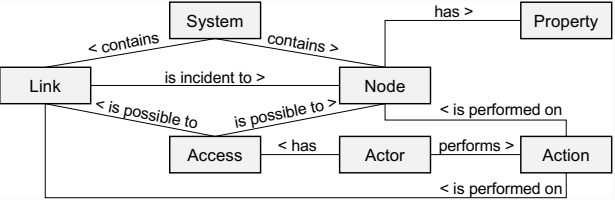
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└─TIM: three families of security models

1:05–1:40. Three sources, then the name.
One of these research efforts is the TIM meta-model, proposed by Zoltán. TIM stands for “Time Is Money”.
What makes it useful is that it combines three families of security models that normally live apart: the graph structure of attack graphs, the probabilistic events of stochastic models, and the attacker–defender framing of game theory.



What TIM models



Nodes and links

servers, routers, connections

Attackers: maximise gain

Defenders: minimise damage

Actions

$$a = (\varphi_a, \psi_a, c_a, d_a, p_a)$$

precondition, postcondition,

cost, duration, success probability

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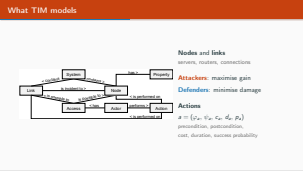
What TIM models

1:40–2:10. Gesture at the diagram; do not read every box.

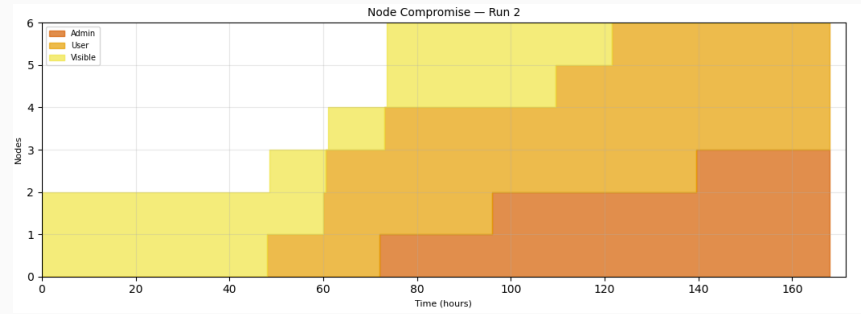
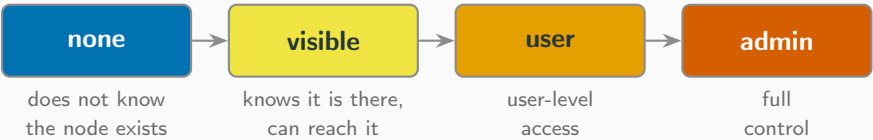
TIM models an organisation's IT assets. The network is composed of nodes and links, which model servers, routers and the connections between them.

There are attackers, trying to make a monetary gain by exploiting the system, and defenders, trying to minimise the damage to it. Both of these actors have a host of actions they can employ to pursue their goal.

An action is a tuple: a precondition that has to hold for it to run, a postcondition describing what it changes, a cost, a duration, and a success probability. Do not read the symbols out — name the five parts.



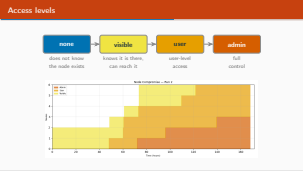
Access levels



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Access levels



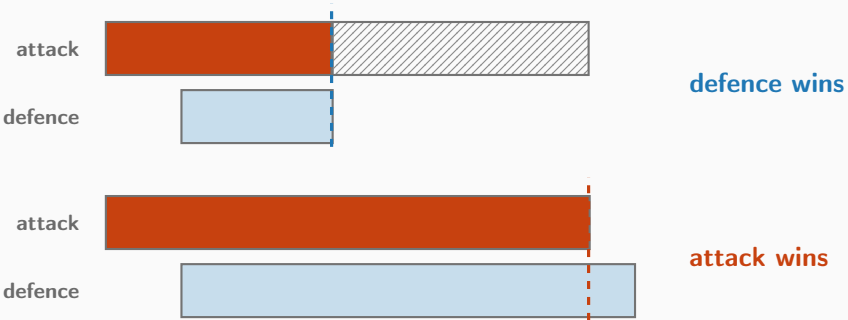
2:10–2:35. These are the same colours they will see in the demo — say so.

The access an actor has to a node is modelled in four levels: none, visible, user and admin, and it is tracked separately for every actor. Gaining privileges on a machine and moving laterally to the next one are both just changes of this access level.

Point at the chart: that is one week of a real run, six nodes climbing from visible, through user, to admin.

Then partial observability: the attacker starts out seeing only the nodes exposed to the internet and has to discover the rest through their own actions. The defender sees the whole network but does not automatically see the attacker, because detection is probabilistic. Both sides act on incomplete information.

Physical time

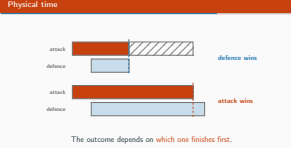


The outcome depends on which one finishes first.

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Physical time

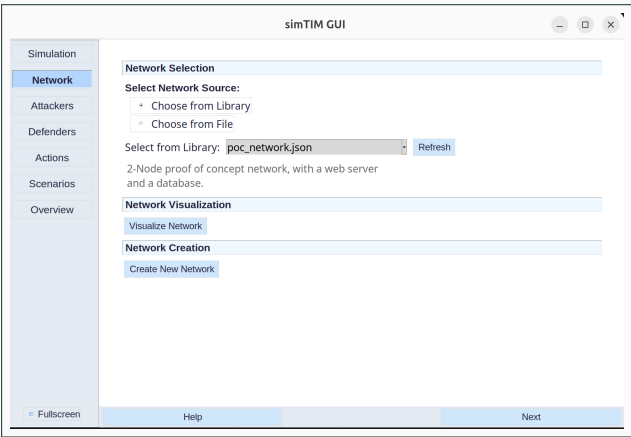


2:10–2:30. Let the picture carry it — two sentences only.
An important addition the TIM meta-model makes is the ability to model physical time. In real life, the duration of an attack and how quickly the defenders can respond is crucial — but former models did not capture this dimension.
Gesture: same attack, same defence, only the durations differ. That is the whole idea, and you will see the number for it at the end of the demo.

simTIM: a full implementation of TIM

```
n1 = Node("web server")
n1.set("framework", ...)
n2 = Node("database")
...
attacker.do(a_tapestry, n1)
```

proof of concept
a *subset* of TIM

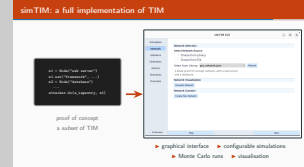


- ▶ graphical interface
- ▶ configurable simulations
- ▶ Monte Carlo runs
- ▶ visualisation

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simTIM: a full implementation of TIM



2:30–3:00. Respectful — the proof of concept did its job.
The original TIM paper came with a small proof-of-concept program, implementing only a subset of TIM’s capabilities.
This is where simTIM comes in. simTIM aims to be a full implementation of the TIM meta-model: a user-friendly graphical interface, easily configurable simulations, Monte Carlo runs, and visualisation along the way.
Then: “instead of talking about it, let me show you.”

Demonstration

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Demonstration

3:00–8:15. Narrate the walkthrough. Do not improvise.

Simulation tab — one week, preset. Stochastic, so not one run but many; three runs for the demo. Detection bias set to *early*: an action is most likely to be detected just after it starts.

Network tab — networks are JSON, saved and loaded from file. Show the network creation tool.

Attacker — how many, which strategy, capacity (how many actions at once), starting budget. Escalation: gain high privilege, then move laterally. Capacity infinite.

Defender — budget and strategy too, but capacity is limited.

Actions — defence capabilities; attackers can be given zero-days to better reflect reality. Mapped to MITRE ATT&CK and D3FEND.

Scenario comparison — compare parameters against each other. Here: different defence durations.

Overview, then run



```
make install  
make run
```

github.com/noahbehrisch/simTIM

Apache 2.0 · **actively developed**

Four places to plug in:

- ▶ networks (JSON)
- ▶ actions (JSON)
- ▶ **strategies** (one class)
- ▶ detection engines (one class)

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└ Open source and extensible

8:15–8:50. Last slide — leave the QR code up during questions.

Apache 2.0, on GitHub, two commands to run it. And say plainly that it is still under active development — that is an invitation, not a disclaimer.

Apache 2.0, on GitHub, two commands to run it.

Four places to plug in. Be honest about the third one: our strategies today are deliberately simple — escalation, greedy, reactive, proactive. Real adversaries are adaptive and coordinated. We built the arena; better players are exactly what we hope the community brings.

Bring your own network, your own actions, your own strategy. Issues and pull requests welcome.

